

CrisisGuard AI — Required APIs & MCP Servers

Complete integration map for Antigravity build · Google Solution Challenge 2025–26

PRD v1.0

Google Cloud Platform APIs (10)

Vertex AI

Vision / Audio / NLP / Anomaly

Google Maps Platform

Maps JS · Indoor API · Directions

Cloud Pub/Sub

Event bus for inter-service async

Cloud SQL (PostgreSQL)

Analytics DB with read replicas,

GKE

Container orchestration with HPA,

Cloud Run

Stateless AI inference endpoints;

Cloud Storage

CCTV snapshots, incident recordings

Cloud KMS

AES-256 envelope encryption for

Cloud Build

CI/CD: lint → test → build image

Cloud Monitoring

Infra metrics, latency alerts,

Firestore Platform APIs (5)

Firestore Auth

Email · Phone OTP · Google · Apple

Firestore

Real-time NoSQL for incidents,

Firestore Data Connect

Sub-100ms sync for Panic Triage

Firestore Realtime DB

Wearable ↔ dashboard bidirectional

FCM

High-priority push; bypass battery

Third-Party APIs (4)

Twilio REST API

SMS alerts for no-app guests;

Implementation Priority Order

- P0 — Critical:** Firestore Auth + FCM + Firestore + Pub/Sub
- P1 — Core Build:** Vertex AI + Maps Platform + Socket.IO + Redis
- P2 — AI Layer:** Gemini 2.0 + Speech-to-Text + Cloud Vision
- P3 — Advanced:** ARCore + Wear OS Tiles + Nearby Connections
- P4 — DevOps:** Cloud Build + GKE + Cloud Monitoring + KMS

AI & Specialized Google APIs (5)

Gemini 2.0 API

Multimodal crisis chatbot; SITREP

Speech-to-Text API

Voice SOS transcription; 125+

ARCore

AR Green Path; plane detection +

Nearby Connections

Mesh networking fallback for offline

Wear OS Tiles API

Persistent emergency info on staff

MCP Servers for Antigravity AI Agents (6)

Firestore MCP

Firestore reads/writes for AI agents; real-time incident context injection

Vertex AI MCP

AI agents invoke vision/NLP/audio models; manage Vertex endpoints

Google Maps MCP

Indoor floor-plan queries; route calculation API calls from agents

GCP Admin MCP

Agents manage Cloud Build triggers, GKE deployments & monitoring

GitHub MCP

Code commits, PR reviews & branch management during development

Twilio MCP

Agents trigger SMS/voice via Twilio for emergency service fallback

24

Total APIs & Integrations

6

MCP Servers for Agents

10

GCP / Firebase Services

5

AI / ML APIs

4

Third-Party APIs